

EquityCoach — System Architecture

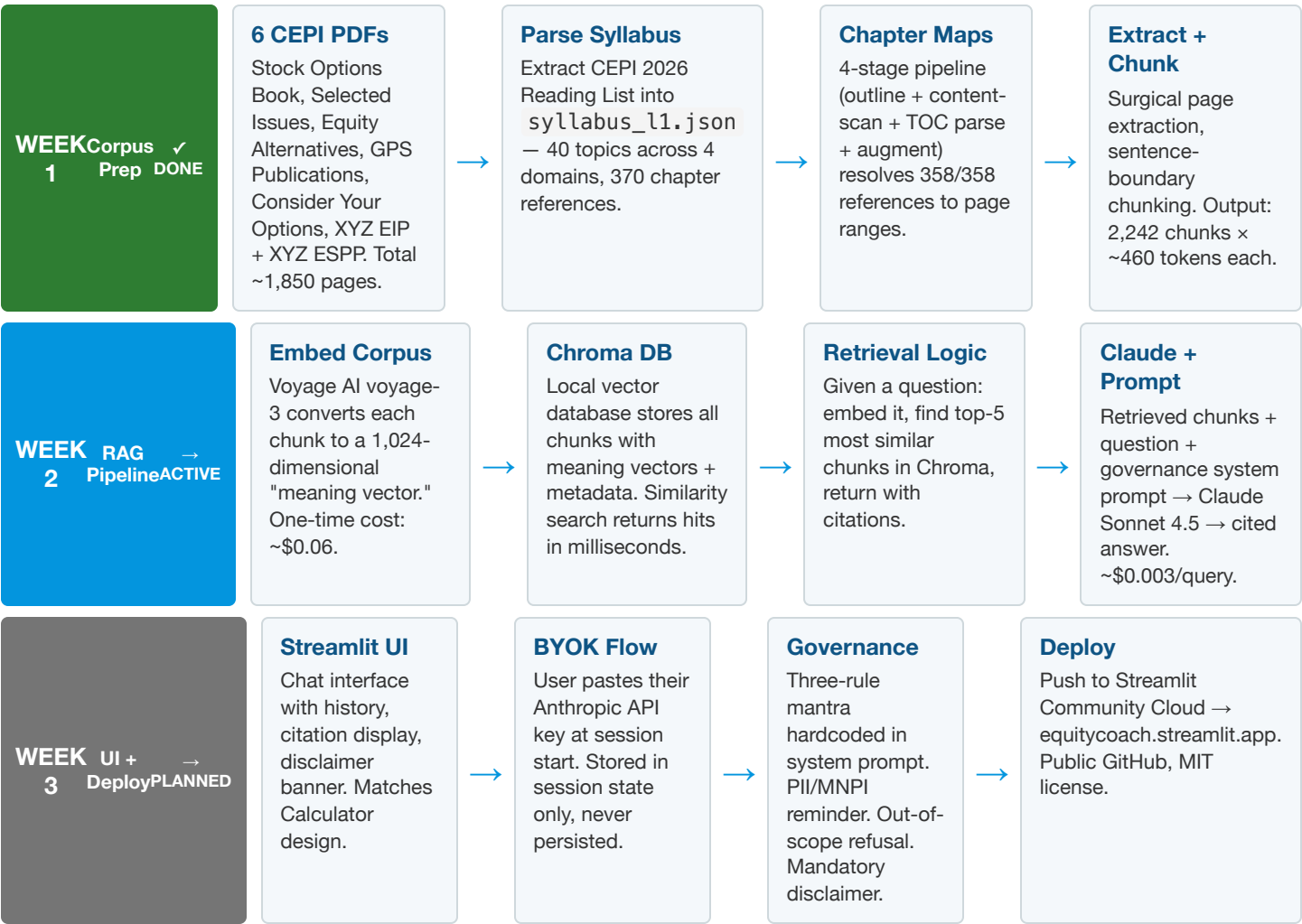
AI Q&A Assistant for CEPI Level 1 Equity Compensation

Priyanka Upreti · Reference document for interview preparation · Launching August 2026

What EquityCoach does: A user asks any CEPI Level 1 question — for example, "When does §16(b) short-swing matching apply to a pre-insider transaction?" — and the Coach returns a plain-English answer with citations to the specific chapter and section in the source material. No hallucinations. Every answer traces back to a real page.

Why the architecture is what it is: The design directly implements the four patterns that shaped every serious case study at NASPP AI in Equity Compensation Week 2026 — citation-based responses (Reddit's Sharelock Holmes), dual-mode agent design (Atlassian's Vestie), prompt-engineering guardrails (Wealthlane's Adhikari formula), and three-rule governance (EY's Goel panel).

The 3-Week Build Pipeline



What Each Piece Does

Voyage AI (Embeddings)

Voyage AI converts text into "meaning coordinates" — a list of ~1,024 numbers that represent what a passage is *about*, not just what words it uses.

Similar meanings → close coordinates. Different meanings → far apart. That's what lets us search by **meaning, not keywords**. Anthropic officially recommends Voyage as the embedding partner for Claude.

Claude (Generator)

Claude Sonnet 4.5 takes the top-5 retrieved chunks + the user's question + a governance-heavy system prompt, and produces the final plain-English answer with mandatory citations.

Same model NASPP AI Week presenters (Baker Tilly, Databricks) used in their demos. Cheapest reliable RAG generation.

Chroma (Vector Database)

Chroma stores all 2,242 chunks along with their meaning coordinates and metadata. When we ask it "which chunks are closest in meaning to this question?", it returns hits in milliseconds via similarity search.

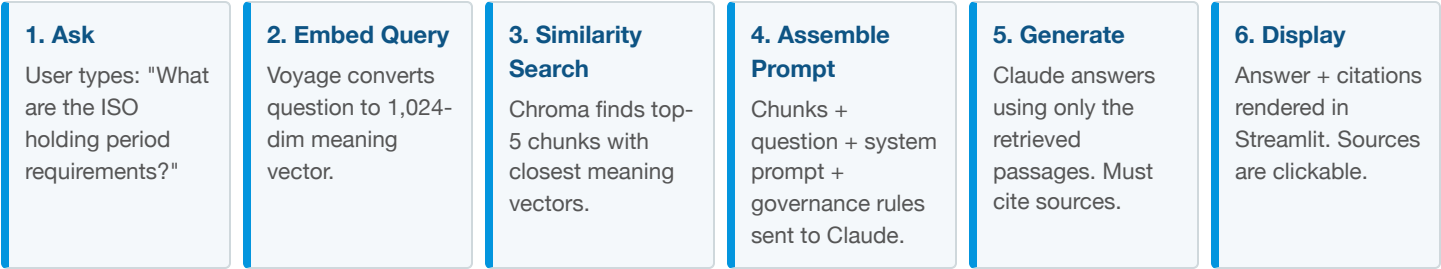
It's free, open-source, runs on the local machine (no cloud dependency). Same purpose as Pinecone or Weaviate; simpler for MVP.

Streamlit (UI + Hosting)

Streamlit is a Python framework that turns Python code into a web app with minimal effort — same stack as the EquityComp Calculator (Project 1).

Deployed to Streamlit Community Cloud's free tier, so hosting cost is \$0. Users bring their own Anthropic API key.

Runtime Flow — What Happens When a User Asks a Question



Latency + cost per query: ~2–4 seconds total. ~\$0.003 per query at Sonnet 4.5 pricing. BYOK model means the user pays this out of their own Anthropic account, not Priyanka. 100 questions = ~\$0.30.

The Governance Layer (Where Real Differentiation Lives)

Every session at NASPP AI Week made the same point: **the AI capability is table stakes; the moat is the guardrails.** EquityCoach hardcodes four layers of governance directly into the system prompt:

1. EY's Three-Rule Mantra

- 1. No sensitive information in unapproved tools
- 2. No important answer without verification
- 3. No consequential action without human authority

These are literally repeated in the system prompt and shown as a footer on every response.

2. Citation-Based Response (Reddit Pattern)

Every claim must include a citation to the specific book and chapter. If Claude tries to answer without a citation, the response is regenerated. This is exactly what Reddit's Sharelock Holmes chatbot does — a hardcoded rule that the response must include Confluence source pages.

3. Out-of-Scope Refusal

If retrieval returns chunks that don't actually address the question, Claude is instructed to refuse rather than guess: "I don't have material on that specific topic in the L1 corpus. Consult a licensed practitioner." Better to say "I don't know" than to hallucinate.

4. Mandatory Disclaimer

Every response begins with: "Educational Q&A over CEPI Level 1 reference materials. Not personal tax, accounting, or legal advice. Consult a qualified professional for situation-specific guidance." Non-negotiable — matches how NASPP AI Week speakers framed their production chatbots.

Cost Model

Component	Type	Cost	Who Pays
Voyage AI embeddings	One-time (corpus prep)	~\$0.06	Priyanka (one-time)
Chroma database	Storage	\$0 (local, free)	—
Streamlit Community Cloud	Hosting	\$0 (free tier)	—
Claude API queries	Per-query	~\$0.003	Each user (BYOK)

GitHub repo	Public open source	\$0	—
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Total to build: ~\$0.06. **Total to host:** \$0/month. Users bring their own Anthropic API key so the tool scales for free.

Interview Talking Points — Common Questions

Q: What is RAG?

RAG stands for Retrieval-Augmented Generation. Instead of asking the LLM a question and hoping it knows the answer from its training data, you first retrieve the most relevant passages from your reference documents, then feed those to the LLM as context. The LLM's answer is grounded in what it retrieved. It's the pattern behind every serious production AI chatbot in equity comp — Atlassian's Vestie, Reddit's Sharelock Holmes, and the case studies Baker Tilly presented at NASPP AI Week.

Q: Why did you build this instead of just using ChatGPT?

Two reasons. First, ChatGPT will hallucinate — it'll confidently produce a wrong IRC citation or wrong section number and no one can tell without checking the book. RAG grounds every answer in the actual reference material and cites the source. Second, ChatGPT doesn't know equity compensation deeply. Even when it's right, it doesn't know which section of the CEPI textbooks the answer comes from. My tool is deliberately narrow — it only answers L1 exam questions, and it cites the exact passage.

Q: How do you prevent hallucinations?

Two mechanisms. First, the system prompt requires every claim to include a citation to a specific book and chapter — if Claude tries to answer without one, the response gets flagged. Second, if the retrieval step returns chunks that don't actually address the question, Claude is instructed to refuse rather than guess. Same three-rule framework Abhinav Goel from EY laid out on Day 3 of NASPP AI Week: no important answer without verification.

Q: Who pays for the API calls?

Each user brings their own Anthropic API key — the BYOK model. It costs about \$0.003 per question at current Claude Sonnet 4.5 pricing. 100 questions costs a user about 30 cents. This keeps the tool free to host and free for me to maintain regardless of usage volume.

Q: Isn't this just a wrapper around a language model?

The value isn't in the model call — it's in the retrieval and governance layers. Getting the corpus scoped correctly (only L1 exam material, no L2/L3), chunked correctly, tagged with rich metadata, and stored so retrieval surfaces the right passages is where the engineering effort went. The governance layer — citations, refusals, disclaimers — is what makes it safe to use in a compliance-sensitive field. That's the difference between a demo and something a practitioner would actually rely on.

Q: What's next after the L1 corpus?

Two directions. Deeper: extend to Level 2 and Level 3 CEPI material once I pass Level 1 in November. Broader: apply the same pattern to reconciliation workflows. That's the Vest-to-Payroll Reconciler I've been designing as a follow-up project — same RAG-plus-governance architecture, but the "corpus" becomes an equity plan admin's SOPs, control matrices, and prior variance analyses. The pattern generalizes.

Q: What's the biggest technical challenge you've hit?

Corpus prep, not the AI part. Three of the five textbook PDFs had broken or missing outlines. I had to write a chapter-mapping pipeline that combined outline parsing, content scanning, printed TOC parsing with offset detection, and manual augmentation — four stages to hit 100% syllabus resolution. That's the least glamorous part of the project but the most important — if the corpus mapping is wrong, retrieval is wrong, and every answer is wrong. Got it to 358/358 references resolved.

Tech Stack Quick Reference

Layer	Choice	Why
Language	Python 3.11	Same as EquityComp Calculator; wide equity-comp familiarity
RAG Framework	LlamaIndex	Purpose-built for RAG; less boilerplate than LangChain
Embeddings	Voyage AI voyage-3	Anthropic-recommended; \$0.06/1M tokens
Vector Store	Chroma	Free, local, no cloud dependency
LLM	Claude Sonnet 4.5	Best price/performance; used in demos at NASPP AI Week
UI Framework	Streamlit	Consistent with Calculator; one-click deploy
Hosting	Streamlit Community Cloud	Free tier; zero DevOps
Auth Model	BYOK (Bring Your Own Key)	Users pay their own API costs; no billing infrastructure

Public Assets

- **GitHub repo:** github.com/priyanka-upreti/equitycoach — Week 1 code shipped, MIT license, public commit history
- **Live URL:** `equitycoach.streamlit.app` (reserved; launching August 25, 2026)
- **Newsletter Issue 1:** "AI in Equity Comp Is Already Here — 6 Things I Learned at NASPP AI Week 2026" — establishes the design context for anyone who wants to know what shaped the Coach's architecture

One-Line Summary for the Elevator Pitch

"EquityCoach is a retrieval-augmented AI Q&A assistant for CEPI Level 1 exam prep. It uses Voyage AI to embed the entire L1 reference corpus into a Chroma vector database, retrieves the most relevant passages for each question, and asks Claude to answer using only those passages with mandatory citations. Same architectural pattern Reddit and Atlassian use for their internal equity comp chatbots — just applied to exam prep instead of employee support."